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Data-Driven Optimization for Oilfield Operations

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Abstract

Field operations and logistics optimization in the oil and gas industry are complex by nature. They involve high-risk dynamic variables and significant resource limitations. This paper introduces two interrelated AI-driven optimization models designed to increase operational efficiency and decision-making in field operations and consumption tank trucking.

The first model focuses on increasing worker efficiency using advanced prioritization techniques. It models behavioral study data and iterative reinforcement learning to improve task planning and execution. Key innovations include Fibonacci prioritization for task scoring, K-means clustering for spatial analysis, and creating a Prophet model to predict production trends and anomalies. Rigorous testing with simulations demonstrates the model's ability to maximize production output, reduce costs, and increase safety by overcoming challenges such as parameter tuning and managing large datasets through the consolidation of data sources.

The second model addresses consumption tank trucking optimization using OptaPlanner (Geurts et al. 2020), an open-source framework that uses heuristic algorithms such as local search, simulated annealing, and Tabu search. This solution improves route planning, task prioritization, and resource allocation for specific domain-related constraints like tank rates and delivery schedules. The model aims to dispatch and deliver consumption-based liquids like methanol, which are key to ensuring smooth operations in the field, such as preventing pipelines from freezing. The model is also enhanced by reinforcement learning techniques for dynamic improvements over time. Deployment through the Java Spring Boot framework on AWS ensures scalability and integration with real-time data input.

Together, these models demonstrate the potential of combining reinforcement learning methods and predictive analytics in oil and gas to optimize real-time inputs and resource utilization in this high-risk environment.

Introduction

Efficient operations and logistics management are critical for an industry where safety, compliance, and productivity meet resource-intensive supply chains. The complexity of field operations and oil and gas transportation logistics is further exacerbated by the ever-changing environment, fluctuating resource

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demands, and a high risk of errors in operations. Traditional methods rely on manual processes and static models, which struggle to efficiently manage the large permutations of operations and resource allocation.

This paper introduces two AI-powered optimization solutions designed to address these challenges. The first focuses on field operations, providing a real-time framework that prioritizes tasks, increases route efficiency, and improves decision-making through feedback-driven machine learning techniques. It also helps field operators work more safely by deducing the best sequence of action items from millions of combinations.

The second solution addresses logistics management for consumption liquid trucking. It leverages OptaPlanner (Sniedovich 2015) to automate distribution routes, increase efficiency, reduce costs, and lower greenhouse gas emissions. This model tackles scalability and computational efficiency to ensure a consistent supply for both small and large businesses at scale. Critical functions benefit from streamlined logistics.

Statement of Theory and Definitions

The proposed optimization models integrate advanced heuristic, metaheuristic, and machine learning techniques, based on the principles of constraint programming and iterative improvement, to address challenges in field operations and logistics management.

Key theoretical components include:

Fibonacci Prioritization: A novel scoring system that weights tasks based on urgency, external factors like weather, SCADA data (Gungor and Hancke 2009), and spatial dynamics using a distance matrix.

K-means Clustering: A machine learning technique to group spatially proximate wells, reducing computational overhead during route optimization.

Prophet Modeling: A forecasting algorithm for predicting production trends and anomalies, facilitating proactive decision-making.

Traveling Salesman Problem (TSP): A classic routing problem applied for optimal sequencing of tasks across well locations.

Constraint Programming: Enforces delivery timelines, tank capacities, vehicle availability, and refilling urgency in the logistics model.

Heuristic and Metaheuristic Algorithms: Techniques such as Local Search, Simulated Annealing, and Tabu Search are used to optimize scheduling and routing under dynamic conditions.

Reinforcement Learning and Greedy Algorithms: Integrated for iterative refinement, allowing the models to adapt dynamically to changing variables and improve over time.

SCADA Data: Real-time operational data sourced from Supervisory Control and Data Acquisition systems. (Baek et al. 2020)

Distance Matrix: A mathematical framework representing distances between multiple locations, crucial for route and task optimization.

EZOps API: A data source for real-time tank rates and operational parameters.

Hyperparameters: Tunable parameters in machine learning models that influence performance and accuracy.

By combining these elements, the models enable dynamic, adaptive optimization of tasks and routes, addressing industry-specific challenges in operational planning and resource logistics.

Description and Application of Equipment and Processes

The optimization models use a combination of advanced data preparation, heuristic modeling, routing optimization along with forecasting, and iterative refinement techniques, used collectively to meet the operational and logistical challenges in the oil and gas sector, as shown in Fig. 1.

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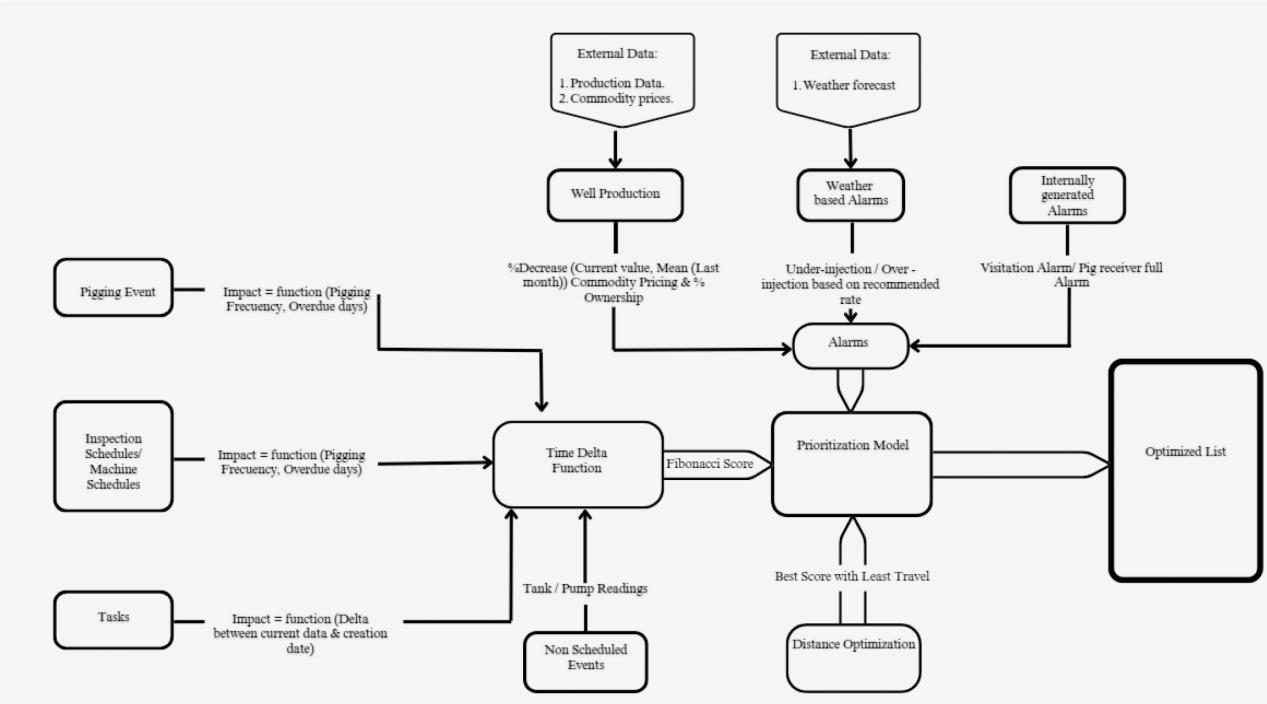

Figure 1—Figure shows all the inputs provided to the prioritization model. The inputs include operational activities passed through a time delta function and internally generated alarms that are given a Fibonacci score and then optimized for distance to create an optimized list.

Data Labeling and Ingestion

Data Preparation: Tasks are prioritized using Fibonacci prioritization (Lindstedt and Jones 2016), considering factors like urgency, weather conditions, SCADA sensor readings, and spatial distances. This ensures that critical tasks are flagged for immediate action.

Data Sources: Real-time data is ingested via the EZOps API, providing information on tank rates, fleet details, and operational metrics essential for logistics optimization.

Optimization Processes

Heuristic Data Modeling: The models integrate experience-based rules with advanced algorithms to address dynamic operational variables. Heuristics are essentially rule-of-thumb concepts derived from experienced operators and translated into logic.

Routing Optimization: K-means clustering is used to group wells into manageable clusters, feeding into a Traveling Salesman algorithm for determining optimal routes.

Forecasting: Prophet modeling predicts production declines and flags anomalies, ensuring tasks are prioritized to mitigate risks.

Constraint Programming: Logistics models handle constraints like tank capacities, delivery timelines, vehicle availability, and refilling urgency.

Infrastructure and Deployment

Platform: The logistics model is developed using OptaPlanner and deployed via a Java Spring Boot framework. AWS FlightControl ensures scalability and efficient handling of high-demand conditions.

Automation: CRON jobs are scheduled for periodic updates, ensuring real-time adaptability to changing variables.

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Testing Framework

Simulations: Both models are tested using simulated data—the simulation is modeled for three months for field operations and various logistics scenarios for consumption tank trucking. Scenarios include high-demand conditions, route changes, and varying operational constraints like extreme weather and Ad hoc production issues.

Performance Metrics: Evaluation focuses on KPIs like reduced operational costs, improved task prioritization accuracy, delivery efficiency, cost per litre, and risk mitigation.

Iterative Refinement and Feedback

Alarming System: An integrated system triggers alerts for high-priority tasks, minimizing production downtime. An example of this is a data point of the hourly condensate production value of a well that is compared to its historic average to flag if it is below the accepted threshold.

Field Feedback: Continuous improvement is achieved through field technician inputs on task relevance, route feasibility, and operational outcomes. User feedback loops are critical for refining the models post-deployment. A feature called 'Early operator Intervention' also allows the user to dictate action items that cannot be skipped on a certain day due to field conditions that cannot be predicted by data in the system allowing the operators to add human intuition as an input.

Solver Tuning: Iterative optimization adjusts hyperparameters and solver configurations to enhance performance dynamically. Parameters like time to complete on certain action items or how drastically their priorities hike as a function of time can be adjusted through configurations.

Case Study: The case study was held in an active oilfield being operated by two operators with task description including inspections, pigging, production monitoring of oilfield operational conditions, and the readouts of equipment under different oilfield conditions. For a month, their performance was evaluated with and without the optimization system in place with comparisons made in terms of efficiency, safety, and production impact.

The team of operators and their behavior were closely tracked by recording their actions in the field data capture system (EZOPS). This made it possible to gather data on what action item was completed in which field and at what priority. The system also had data on all of the action items due on a day and the ones that the operators picked as a priority. This data was key in understanding the behavior difference in making operational micro decisions with and without the optimization. Previously, the operators would complete a critical action item like sending a pig through the pipeline to avoid clogging and freezing the pipeline and then move on to another location with critical items. The optimization, on the other hand, continuously synthesizes data on all possible actions to be completed on a field and recommends a proactive action that would be due in a few days to be completed in the same location, before moving on. This would prevent the operators from revisiting that location again in a few days and optimize the route for the operators. The optimization also recommends taking micro-decisions that do not seem as urgent but eventually trickle down to running a safer operation. An example would be collecting tank levels and recording the readings on tanks and equipment. These readings ensure a smoother auditing process for the field leads as well as ensure the right data is available for the cross-shift to take over the field workload smoothly. The tank levels would be used by the optimization to recommend an optimized trucking load for when all the tanks in the field need a top-up. The optimization also continuously synthesizes hourly well production data and predicts the outlier behavior of a well that will prevent it from making its expected production for the day. The operators were able to identify production issues sooner and mitigate them faster. The onus on the operators to read complex SCADA charts before starting a shift was also lifted. Decisions like these eventually compounded and solid improvements in the KPIs were observed after the field team started referring to the recommended route scheduling by the optimization.

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Presentation of Data and Results

The optimization models deliver significant improvements across various key performance indicators (KPIs), highlighting their effectiveness in enhancing operational efficiency, cost savings, and safety in field operations and logistics management. This study compares the performance of a real-world field operation team consisting of both green and brown operators (varying field experience) before and after optimization. The system recorded each task that was performed, the time to complete and the effectiveness of their schedules on important metrics like total flow of production from the well, number of production issues, number of delinquent items, and so on.

Field Operations Results

Efficiency: The operator optimization model increased the route efficiency (actions completed per hours billed) by 31%, allowing technicians to dedicate more time to task execution.

Safety: Delinquent incidents are reduced by 61%, attributed to improved task prioritization and adherence to safety protocols.

Production Uptime: Forecasting methods identify potential production issues earlier, decreasing downtime by 47%.

Logistics Management Results

Cost Efficiency: The logistics optimization model demonstrates cost savings through reduced delivery times and improved truck utilization. There was a 20% reduction in the landed cost per lire.

Delivery Efficiency: Average delivery distances are minimized, resulting in consistent supply chain performance even under fluctuating demand conditions. There was a 35% increase in load utilization impacting the overall efficiency of the delivery.

Environmental Impact: Optimized routes contribute to reduced emissions, aligning with sustainability goals.

Conclusion

The optimization model presented in this study shows significant potential to revolutionize field operations and logistics management in an industry characterized by complex, dynamic challenges.

AI-based field operations models increase efficiency by efficiently prioritizing operations, improving route efficiency, and adapting to changes in real time. Integration with forecasting tools and warning systems allows for proactive decision-making, reducing downtime and improving safety. Challenges such as data integration and computational requirements remain, though continuous improvements and advancements in scalability indicate the broad applicability of the model in a wide range of operating situations.

In the same way, the logistics optimization model demonstrates that meta-heuristic methods can handle real-world complexity efficiently and effectively. Using a scalable framework ensures that the model adapts to increasing operational demands.

Together, these models offer a comprehensive approach to optimizing operations and logistics ensuring that field operations are safer, greener, and cost-efficient.

Abbreviations

API : Application programming interface
AWS : Amazon Web Services
CRON : Command Run ON. A process scheduler that executes commands at specified times.
EZOPS : Oil and Gas management software
KPI : Key Performance Indicator

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SCADA : Supervisory Control and Data Acquisition

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Appendices

Scalability and Infrastructure

The operator optimization model requires robust computational infrastructure, including high-capacity servers and efficient data storage solutions. Clustering methods enhance scalability, enabling the model to handle datasets for clients with over 1,000 wells.

The logistics optimization model is designed for scalability and high-performance demands. Key infrastructure components include:

AWS Deployment: Leveraging AWS FlightControl ensures seamless scalability and reliability, even during peak operational loads.

Solver Efficiency: OptaPlanner's ability to process large-scale routing and scheduling scenarios is enhanced through domain-specific configurations and solver optimizations.

Data Handling: Integration with EZOps API provides real-time updates on tank rates and fleet availability, supporting dynamic adaptability.

User Configuration

Interactive Dashboard: Users can adjust delivery priorities, fleet assignments, and routing constraints via a customizable web-based interface.

Dynamic Updates: Scheduled CRON jobs enable periodic updates to reflect real-time changes in logistics data.

Future Directions

Reinforcement Learning Enhancements: Integration of advanced reinforcement learning algorithms to further refine task prioritization and routing adaptability.

Expanded Applications: Extending the model's applicability to other resource-intensive logistics networks, such as water or chemical transportation.

Sustainability Metrics: Adding modules to measure and optimize environmental impact, such as emissions tracking and fuel efficiency analysis.

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